PROJECT REPORT

Food Ordering Behaviour and Consumer Trends:
A Structured Analysis of Choices and Habits

Team ID: SWTID-2026-2426

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Date: 18 July 2026

1. Introduction

1.1 Project Overview

The food delivery industry has grown highly competitive, and businesses increasingly struggle to understand customer behaviour, optimise delivery performance, and maximise revenue even though large volumes of order data are already available. This project, “Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits,” addresses that gap by analysing food ordering data using Tableau to uncover meaningful patterns in how, when, and why consumers order food.

The dataset used for the analysis includes key attributes such as Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City, and Time Taken for Delivery. Interactive Tableau dashboards built on this dataset allow the team to explore how different factors impact order volume, customer preferences, and operational performance, surfacing insights such as top-performing cities, high-demand cuisines, peak mealtimes, and delivery time distribution.

1.2 Purpose

The purpose of this project is to convert raw food-ordering data into actionable insight for three groups of stakeholders identified during the research:

- Business Analysts, who need to understand order trends, cuisine preferences, and city-wise demand to support strategic, data-driven decisions.
- Operations Managers, who need visibility into delivery performance and workload distribution to improve efficiency and reduce delays.
- Customers, whose ordering habits (frequency, timing, price sensitivity, platform and payment preference) form the underlying behaviour the analysis is built around.

By structuring the analysis around these needs, the project aims to help stakeholders recognise areas of strong performance as well as concrete opportunities for improvement.

2. Ideation Phase

2.1 Problem Statement

Two customer problem statements were defined to ground the research in real user pain points:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A working professional aged 25–35 who orders food on most weekdays due to a busy schedule	Get a satisfying meal delivered quickly without spending much time deciding	The app shows too many options, delivery times are inconsistent, and the final price often differs from what was shown initially	Menus aren't personalized to past orders, and platforms add delivery/platform fees only at checkout	Frustrated and rushed
PS-2	A budget-conscious college student who orders food occasionally for group meals or late-night cravings	Find the most affordable meal option with the best available deal	It's hard to compare prices, discounts, and delivery fees across different food ordering apps	No single platform shows the true final cost upfront or compares offers across apps	Overwhelmed and unsure about overspending

Figure: Customer Problem Statements (PS-1, PS-2)

2.2 Empathy Map Canvas

An empathy map was built for the project's primary persona — a working professional aged 25–35 who orders food on most weekdays because of a busy schedule (based on Problem Statement PS-1).

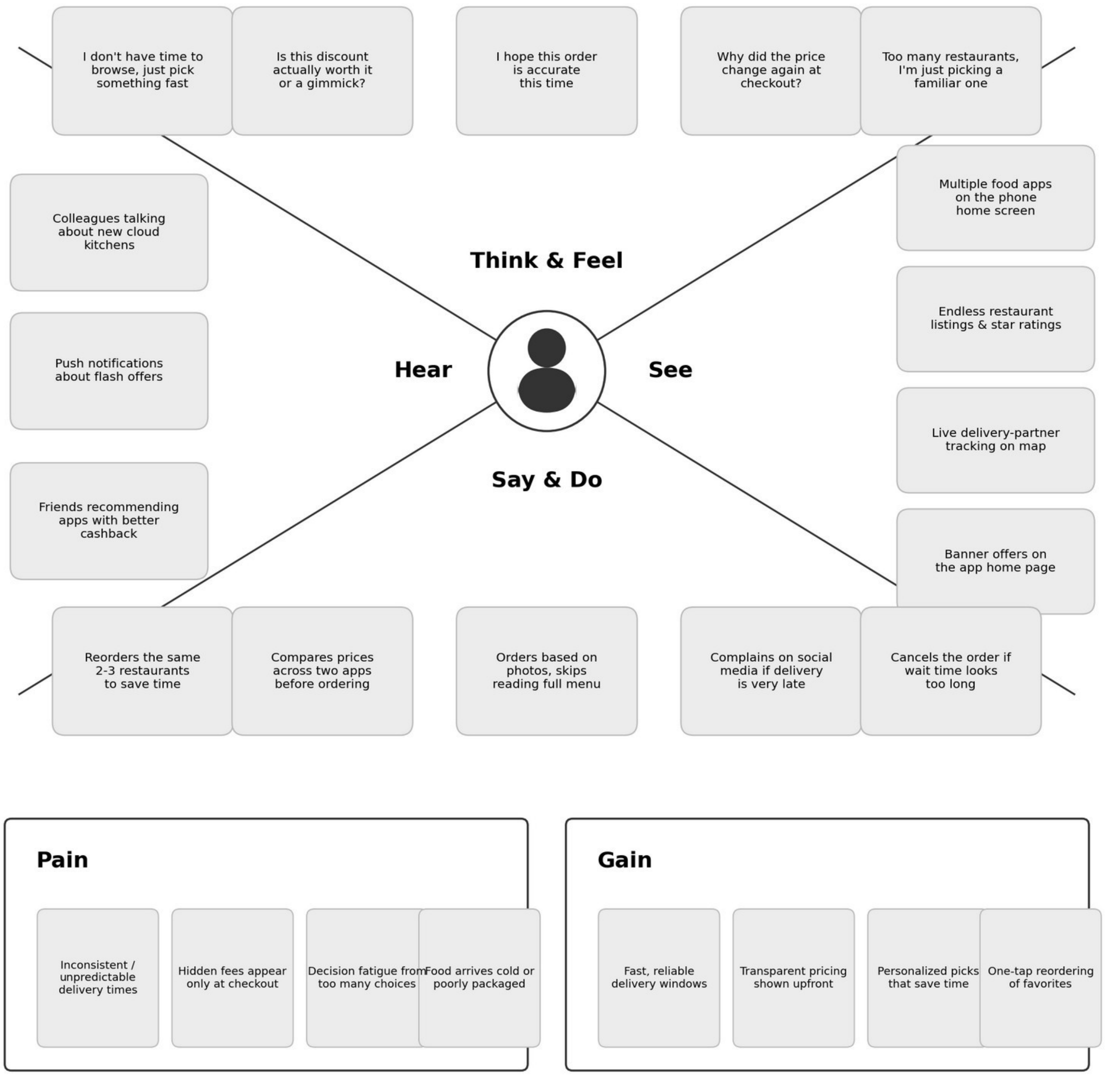


Figure: Team Empathy Map — Food Ordering & Delivery App User

These patterns — valuing speed and transparency, being influenced by word-of-mouth and in-app offers, and being most frustrated by unpredictable delivery times and hidden charges — directly shaped the focus areas of the structured analysis.

2.3 Brainstorming

The team ran a structured brainstorm around the question: "How might we identify and structure the key behavioural patterns, influencing factors, and habits that shape consumer choices in food ordering, in order to build a clear analytical understanding of trends in the industry?" The session followed two ground rules — go for volume, and be specific and visual.

Step 1: Individual Idea Listing

Each team member wrote down ideas addressing the problem statement individually before grouping.

Sri Chaitra	Chandhana
Order frequency: weekly vs monthly habits	Preferred platform: Swiggy / Zomato / Uber Eats

Weekday vs weekend ordering patterns	Subscription usage: Swiggy One, Zomato Gold
Price sensitivity & discount/coupon usage	Payment method preference: UPI, COD, wallet
Influence of online reviews & ratings	Social media & influencer impact on choices
Delivery time expectations & satisfaction	Age-group variation in ordering behaviour
Impact of festive/seasonal offers	Restaurant loyalty & repeat ordering habits
Health-consciousness in menu choices	Impact of weather/mood on ordering decisions

Step 2: Group Ideas

The team clustered the individual sticky notes into related themes and gave each cluster a sentence-like label, then ranked the themes on an importance-versus-feasibility matrix.

Ordering Frequency & Timing Patterns	How often, when, and under what conditions (weekday/weekend, weather, festivals) consumers order food.
Decision-Influencing Factors	Price sensitivity, discounts, reviews/ratings, delivery time, and health-consciousness that shape choices.
Platform & Payment Preferences	Choice of ordering platform and preferred payment methods (UPI, COD, wallets).
Demographics & Social Influence	Age-group variation, social media/influencer impact, and restaurant loyalty behaviour.
Subscription & Loyalty Programs	Usage of subscription plans such as Swiggy One and Zomato Gold, and their effect on ordering frequency and repeat-purchase behaviour.

Step 3: Prioritization

Individually generated ideas were grouped into five themes and placed on an importance-versus-feasibility matrix:

- Ordering Frequency & Timing Patterns — highest priority (core focus)
- Decision-Influencing Factors (price, reviews, delivery time) — highest priority (core focus)
- Platform & Payment Preferences — supporting theme
- Demographics & Social Influence — supporting theme
- Subscription & Loyalty Programs — secondary angle (lower feasibility within the project timeline)

3. Requirement Analysis

3.1 Customer Journey Map

A customer journey map traced the primary persona's experience of discovering, ordering from, and returning to a food delivery app across five phases — Entice, Enter, Engage, Exit, and Extend — covering steps, touchpoints, goals, emotional highs and lows, and improvement opportunities at each stage.

Figure: Food Ordering & Delivery Journey (Entice → Enter → Engage → Exit → Extend)

Positive moments include excitement about a first-order discount and reassurance from a peer's recommendation; negative moments include too many permissions requested at sign-up, hidden delivery/platform fees appearing only at payment, and long waits for a quiet dispute resolution.

3.2 Solution Requirement

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import & Preparation	Import order dataset (CSV/Excel) into Tableau Clean and pre-process data (remove duplicates & nulls)
FR-2	Order & Revenue Trend Analysis	View order volume trends by day/week/month View revenue trends over time
FR-3	City & Cuisine-wise Demand Analysis	City-wise order volume dashboard Cuisine & meal-type demand breakdown
FR-4	Delivery Performance Analysis	Delivery time distribution by city Delivery fee vs. delivery time comparison
FR-5	Customer Behaviour Insights	Peak mealtime ordering analysis Repeat-order & loyalty pattern view
FR-6	Interactive Dashboard & Reporting	Filter by city, cuisine, date range Export / share summary reports

Non-Functional Requirements:

NFR No.	Requirement	Description
NFR-1	Usability	Dashboards should be intuitive so analysts can filter and interpret data without training.

NFR No.	Requirement	Description
NFR-2	Security	Access to the workbook and underlying data restricted via role-based permissions.
NFR-3	Reliability	Dashboard should show accurate, consistent figures whenever the source data is refreshed.
NFR-4	Performance	Visuals should render and respond to filter changes within a few seconds even on the full dataset.
NFR-5	Availability	Published dashboards should be accessible on Tableau Server / Public whenever stakeholders need them.
NFR-6	Scalability	The workbook should continue to perform well as the order dataset grows over time.

3.3 Data Flow Diagram

At a conceptual level, data flows from the raw order dataset (CSV/Excel or a SQL source) into a data-preparation step where duplicates and nulls are removed and fields such as Cuisine, Meal Type, and City are standardised. The cleaned data is then loaded into Tableau, where it is transformed into the dashboards and visualisations described in the Functional Requirements, before being published for Business Analysts, Operations Managers, and other stakeholders to explore and export.

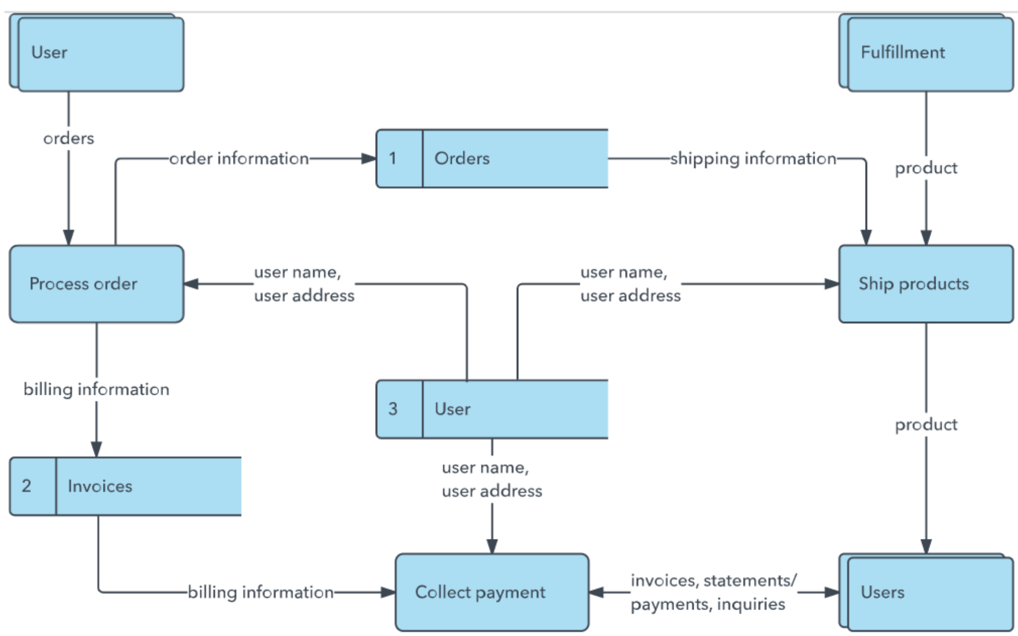


Figure: DFD Level 0 (Industry-Standard Reference) — the numbered-process / data-store notation used to structure the project's own flow: Data Sources → Data Prep → Tableau Desktop → Dashboards → Tableau Server / Cloud.

3.4 Technology Stack

S.No	Component	Description	Technology
1.	Data Source	Raw order-level data with fields such as Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City, Time Taken for Delivery	Excel / CSV / SQL database
2.	Data Preparation	Cleaning, de-duplication, and standardisation of the raw dataset	Tableau Prep / Excel

S.No	Component	Description	Technology
3.	Analysis & Visualization	Building interactive dashboards and charts from the prepared data	Tableau Desktop
4.	Publishing / Sharing	Hosting dashboards for stakeholders to view and interact with	Tableau Public / Tableau Server
5.	Client Access	How stakeholders view and interact with dashboards	Web browser (Chrome / Edge / Safari)

4. Project Design

4.1 Problem – Solution Fit

A Problem-Solution Fit Canvas was used to validate that the proposed data analytics solution genuinely addresses the real problems, constraints, and behaviour of the people it is designed for.

Element	Details
Customer Segment(s)	Business Analysts and Operations Managers within food delivery businesses, and end customers who order food through the platform.
Jobs-to-be-Done / Problems	Analysts need to understand order trends and customer preferences. Managers need to identify delivery delays and inefficiencies. Customers want fast, affordable delivery with good cuisine variety.
Triggers	Increasing competition in the food delivery industry; large volumes of raw order data with no clear way to extract insights; recurring delivery delays and inconsistent service.
Emotions: Before / After	Before: overwhelmed by unstructured data (analyst/manager); frustrated by slow, inconsistent delivery (customer). After: confident, data-driven decision-making (analyst/manager); satisfied and likely to reorder (customer).
Available Solutions	Manual spreadsheet analysis, generic sales reports, and basic dashboards without deep drill-down; decisions often rely on intuition and past experience rather than structured data.
Customer Constraints	Large, complex datasets across many attributes (Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City, Time Taken); limited time to manually process data; no standardized analytical tool.
Behaviour	Analysts and managers manually review raw order and delivery reports. Customers choose cuisine and meal type based on convenience and reorder mainly during peak lunch and dinner hours.
Channels of Behaviour	Online: food delivery app/website, Tableau dashboards accessed via browser. Offline: internal business reports, spreadsheet reviews, in-person operations meetings.
Problem Root Cause	Food delivery businesses collect large volumes of order data but lack a structured analytical approach to convert it into actionable insights, making it difficult to understand demand drivers, customer preferences, and delivery performance.
Your Solution	A Tableau-based analytics solution that visualizes food ordering data (Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City, Time Taken for Delivery) through interactive dashboards — revealing top-performing cities, high-demand cuisines, peak mealtimes, and delivery time trends to support data-driven decisions for analysts and operations managers, and to improve the overall customer experience.

4.2 Proposed Solution

The proposed solution is a set of interactive Tableau dashboards built on the cleaned food-ordering dataset. The dashboards let a Business Analyst or Operations Manager filter by city, cuisine, meal type, and date range to view order and revenue trends, city-wise and cuisine-wise demand, delivery time distribution, and peak mealtime patterns — turning raw order records into the specific insights identified in the Functional Requirements (top-performing cities, high-demand cuisines, peak mealtimes, and delivery time distribution).

Proposed Solution Template

S.No	Parameter	Description
1.	Problem Statement (Problem to be solved)	Food delivery businesses generate large volumes of order and delivery data (Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City, Time Taken for Delivery), but lack a structured analytical approach to convert this data into actionable insights — making it difficult to understand customer behaviour, identify high-demand cuisines/cities, and optimize delivery performance.
2.	Idea / Solution description	A Tableau-based analytics solution that ingests food ordering data and builds interactive dashboards to visualize order volume, cuisine preferences, city-wise demand, peak mealtimes, and delivery time distribution — enabling data-driven decisions for business analysts and operations managers.
3.	Novelty / Uniqueness	Combines multiple operational dimensions (order value, delivery fee, cuisine, meal type, city, delivery time) into a single interactive dashboard rather than siloed reports, enabling cross-functional insights across business, operations, and customer experience from one unified data source.
4.	Social Impact / Customer Satisfaction	Faster identification of delivery delays and service gaps leads to quicker, more reliable deliveries and better cuisine variety for customers, improving overall satisfaction and encouraging repeat orders and platform loyalty.
5.	Business Model (Revenue Model)	The analytics dashboard can be used as an internal BI tool for the food delivery platform, or offered as a subscription-based analytics service (SaaS) to restaurants and delivery partners, with a recurring fee based on dashboard access tiers.
6.	Scalability of the Solution	Built on Tableau, the solution can scale to handle increasing data volume across more cities and cuisines, and can be extended with additional data sources (customer reviews, loyalty data) or automated via Tableau Server/Cloud for real-time dashboard refresh.

4.3 Solution Architecture

Solution architecture is a complex process — with many sub-processes — that bridges the gap between business problems and technology solutions. Its goals are to find the best tech solution to solve existing business problems, describe the structure and behaviour of the software to project stakeholders, define features and development phases, and provide specifications for how the solution is defined, managed, and delivered.

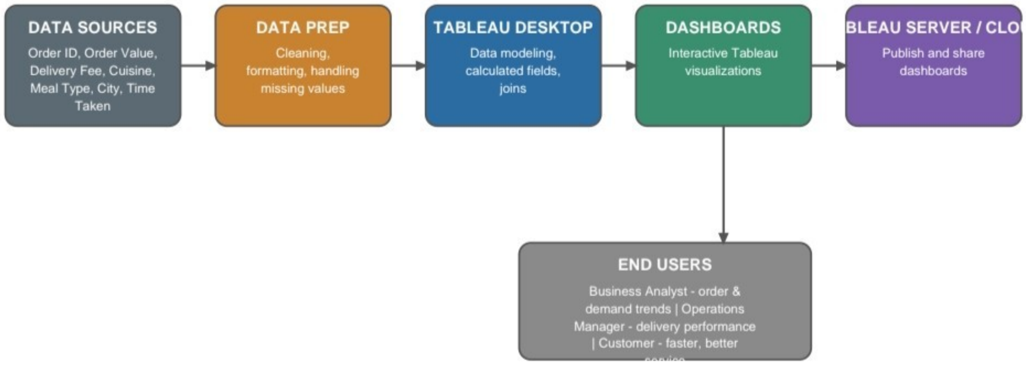


Figure 1: Architecture and data flow of the food ordering analytics solution

The pipeline runs from Data Sources (Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City, Time Taken) through Data Prep (cleaning, formatting, handling missing values), into Tableau Desktop (data modeling, calculated fields, joins), which produces the interactive Dashboards. These are published to Tableau Server / Cloud, from which End Users — Business Analysts (order & demand trends), Operations Managers (delivery performance), and Customers (faster, better service) — draw their insights.

5. Project Planning & Scheduling

5.1 Project Planning

Work was broken into epics and user stories and scheduled across three sprints:

Sprint	Focus	Key Stories
Sprint-1	Foundation & core analysis	Data preparation; order & revenue trend analysis; city-wise demand; delivery performance monitoring; peak-hour workload analysis; dashboard filtering
Sprint-2	Deeper analysis	City drill-down by cuisine; average order value by meal type; delivery fee vs. time comparison; delivery delay reallocation; cuisine preference insights
Sprint-3	Insights & reporting	Peak mealtime insight; summary report export

6. Functional and Performance Testing

6.1 Performance Testing

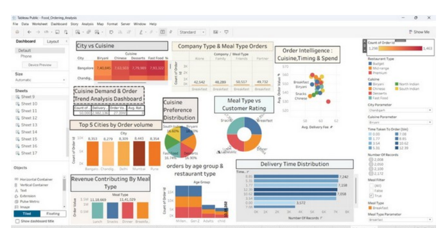
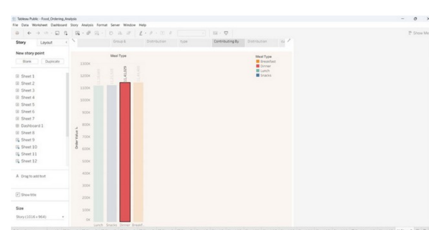
Before publishing, each dashboard was checked against the following expectations:

- Load time: dashboards open and render within a few seconds on the full dataset.
- Filter response: changing a city, cuisine, or date-range filter updates all visuals without noticeable lag.
- Data accuracy: totals and averages shown on each dashboard were spot-checked against the source dataset.

- Cross-device check: dashboards were reviewed in a standard browser window to confirm layout and interactivity hold up outside the Tableau editor.

6.2 Model Performance Testing

The project team documented how the team's actual Tableau Public dashboard and story (“Food_Ordering_Analysis”) for the food ordering behaviour dataset were built, cleaned, filtered, and visualized.

S.No.	Parameter	Details	Screenshot / Values
1.	**Data Rendered**	The Food_Ordering_Analysis dataset was connected and rendered in Tableau Public across multiple worksheets, dashboards, and a story.	17 worksheets (Sheet 1–17) • 2 dashboards (Dashboard 1, Dashboard 2) • 1 story (Story 8) • Key fields: Order Id, City, Cuisine, Meal Type, Restaurant Type, Order Value, Delivery Fee, Age Group, Company/Group Type, Time Taken to Order
2.	**Data Preprocessing**	Categorical fields were standardized into consistent groups so charts and filters aggregate cleanly across the dataset.	Cuisine: Biryani, Chinese, Desserts, Fast Food, North Indian, South Indian • Restaurant Type: Budget, Mid-range, Premium • Meal Type: Breakfast, Lunch, Dinner, Snacks • Age Group: Millennial, Gen Z, Adults, Child • Company Type: Alone, Family, Friends, Partner • Time Taken to Order binned into 8 intervals (0.00–12.39)
3.	**Utilization of Filters**	Parameter and quick filters let users slice every dashboard and story point by location, cuisine, and meal timing.	Restaurant Type • Cuisine • City Parameter • Cuisine Parameter • Time Taken to Order (bin) • Number of Records • Meal Filter (True/False/All) • Meal Type • Meal Type Parameter
4.	**Calculation Fields Used**	Calculated fields power the KPI cards and axis values across the dashboard and story.	Count of Order Id • Avg. Order Value • Avg. Delivery Fee • % of Total Count of Order Id (cuisine share) • Order Value in thousands (revenue by meal type)
5.	**Dashboard Design**	Dashboard 2 combines multiple visualizations with a shared filter panel for interactive, cross-filtered analysis.	 *Live Tableau Public dashboard — 10 visualizations with filter panel*
6.	**Story Design**	Story 8 sequences the dashboard's charts into a guided narrative, moving from demand patterns to revenue and delivery performance.	

S.No.	Parameter	Details	Screenshot / Values
			Story 8 — 9 sequenced story points (full sequence shown as Screenshots 2–10 in Section 7.1)

Tableau Public Dashboard Link: Dashboard 2

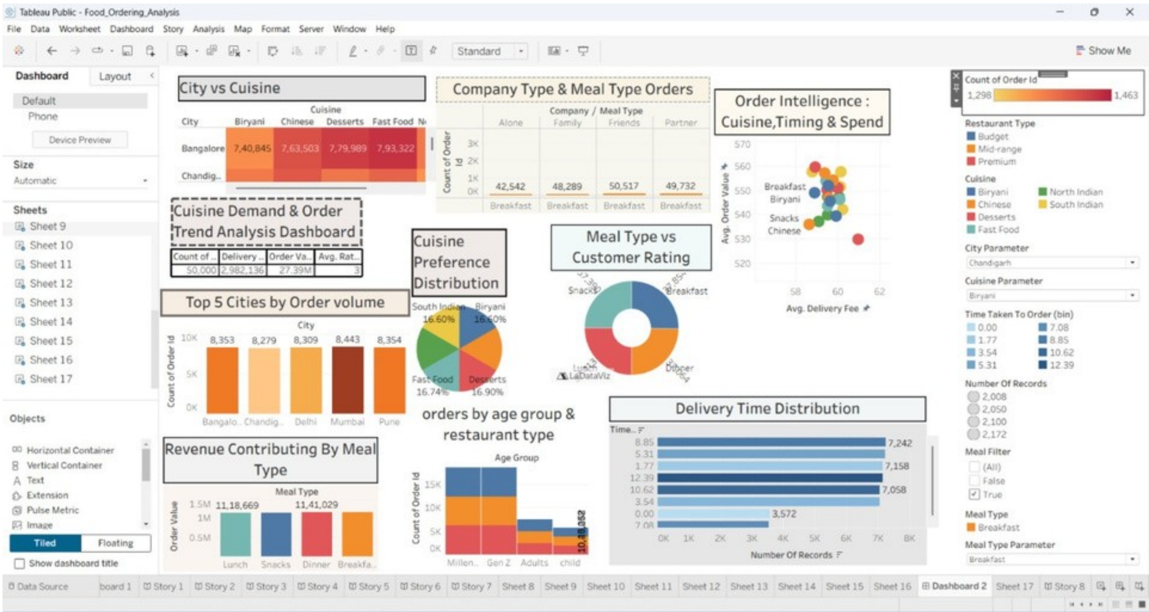
Tableau Public Story Link: Story 8

7. Results

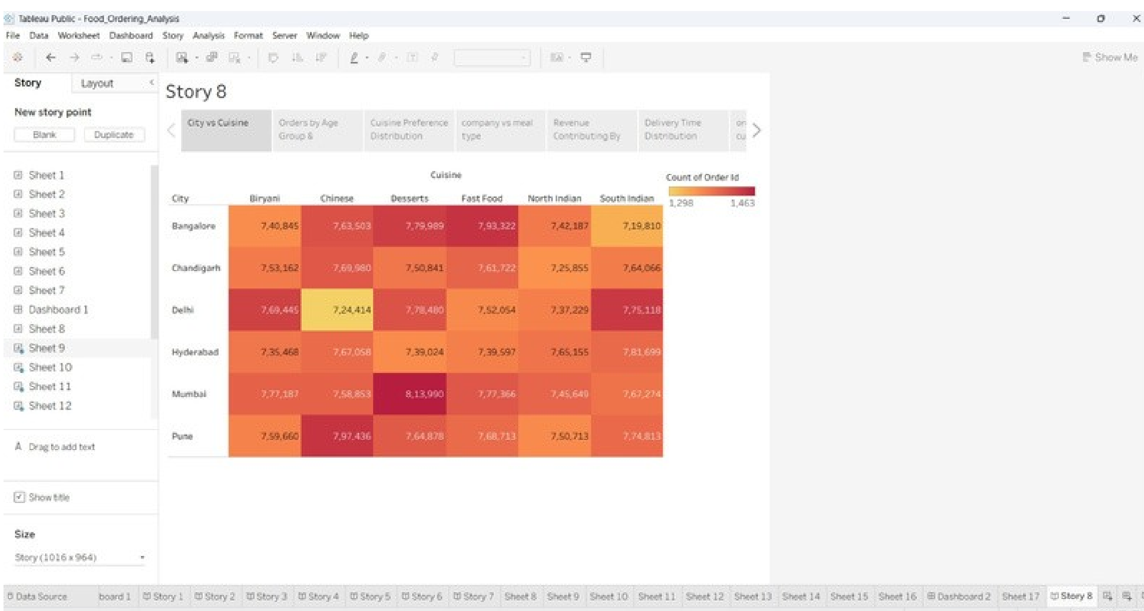
7.1 Output Screenshots

The finalised Food_Ordering_Analysis Tableau workbook — covering the main dashboard, the City & Cuisine-wise Demand view, and the Delivery Performance view — is captured below.

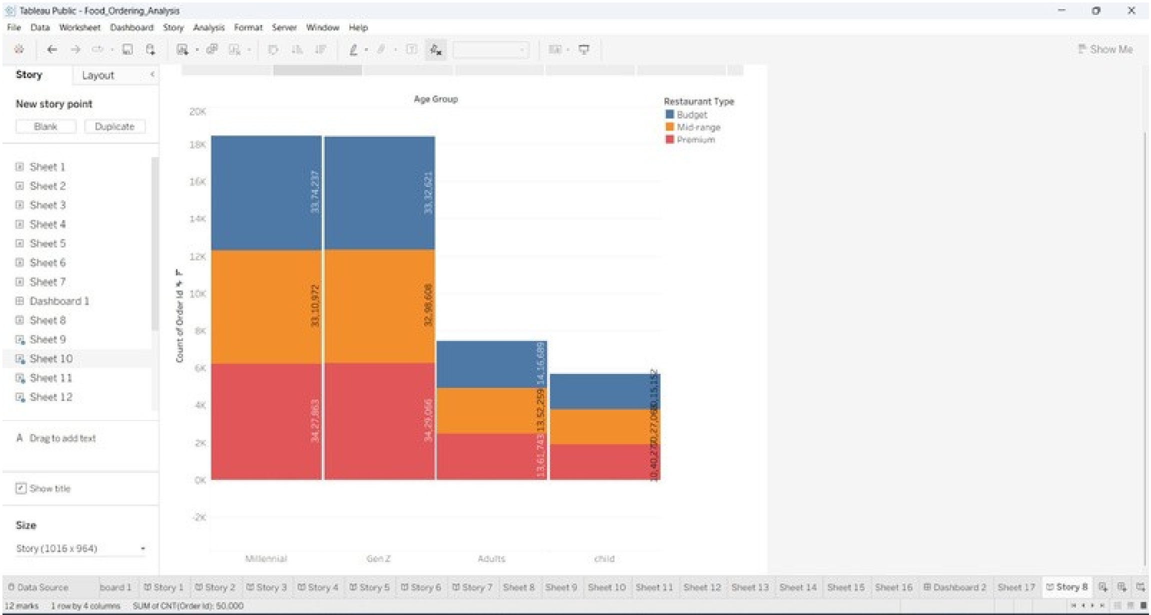
Screenshot 1: Main Dashboard — Food Ordering Behaviour and Consumer Trends



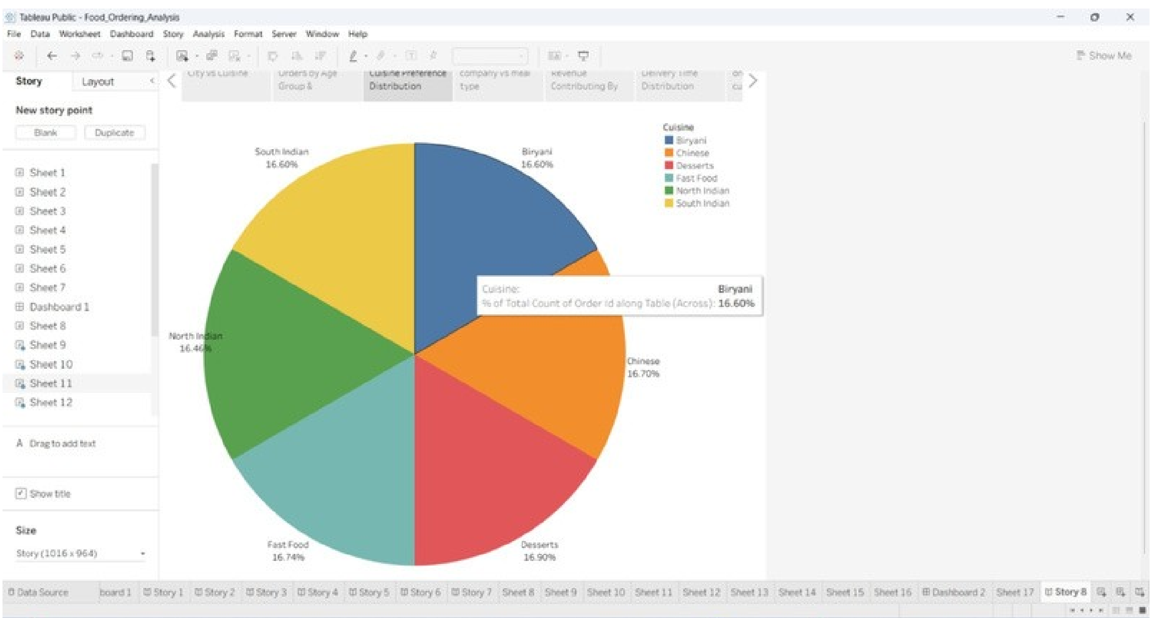
Screenshot 2: City vs Cuisine



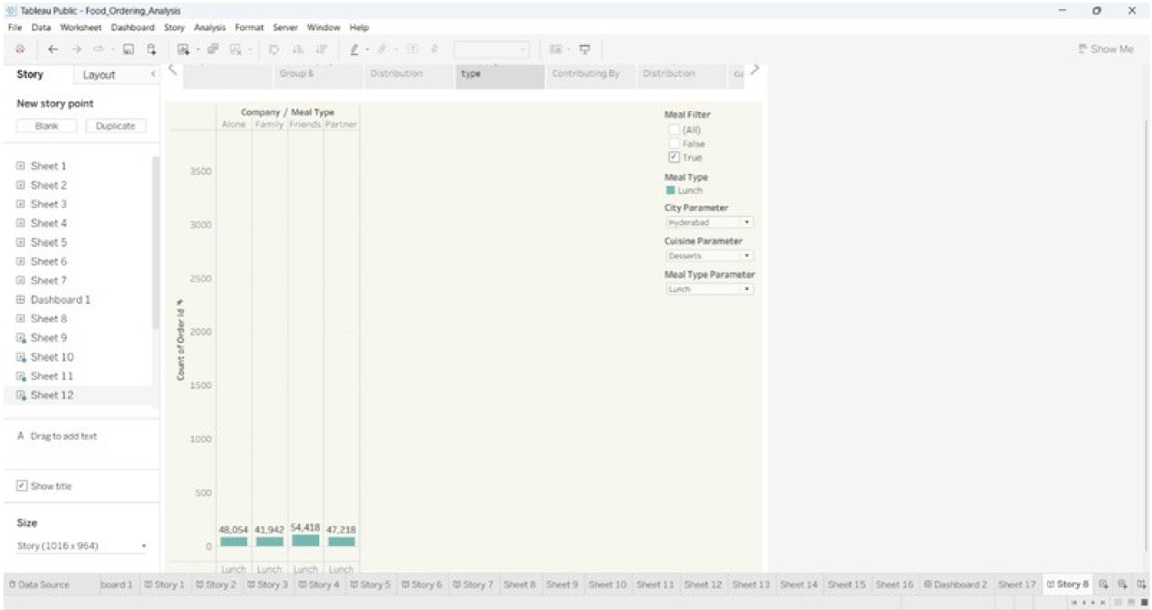
Screenshot 3: Orders by Age Group & Restaurant Type



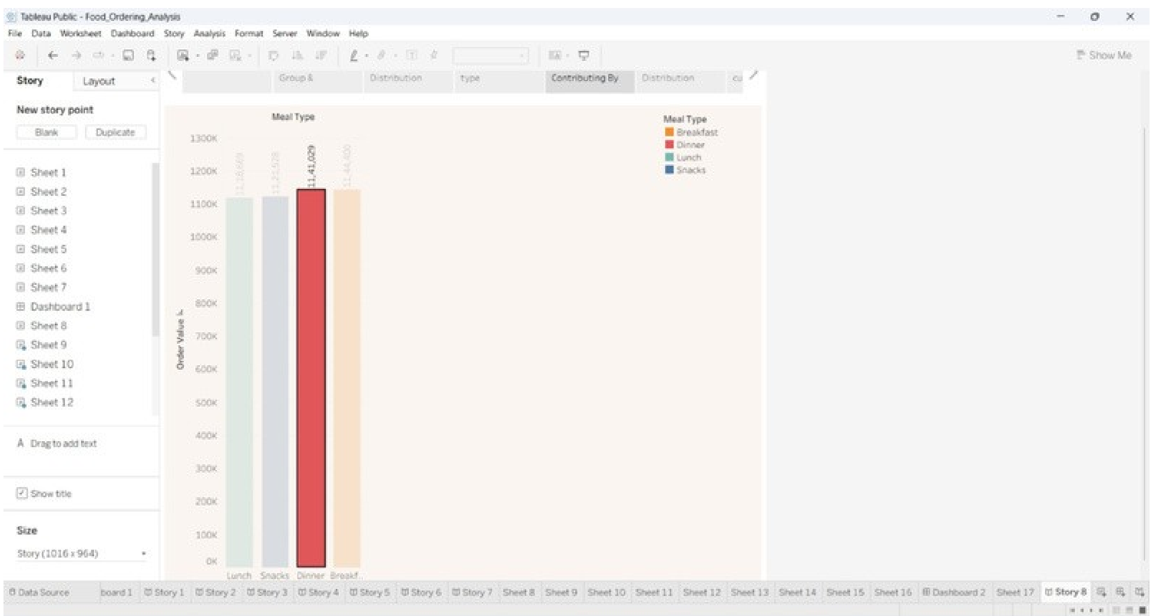
Screenshot 4: Cuisine Preference Distribution



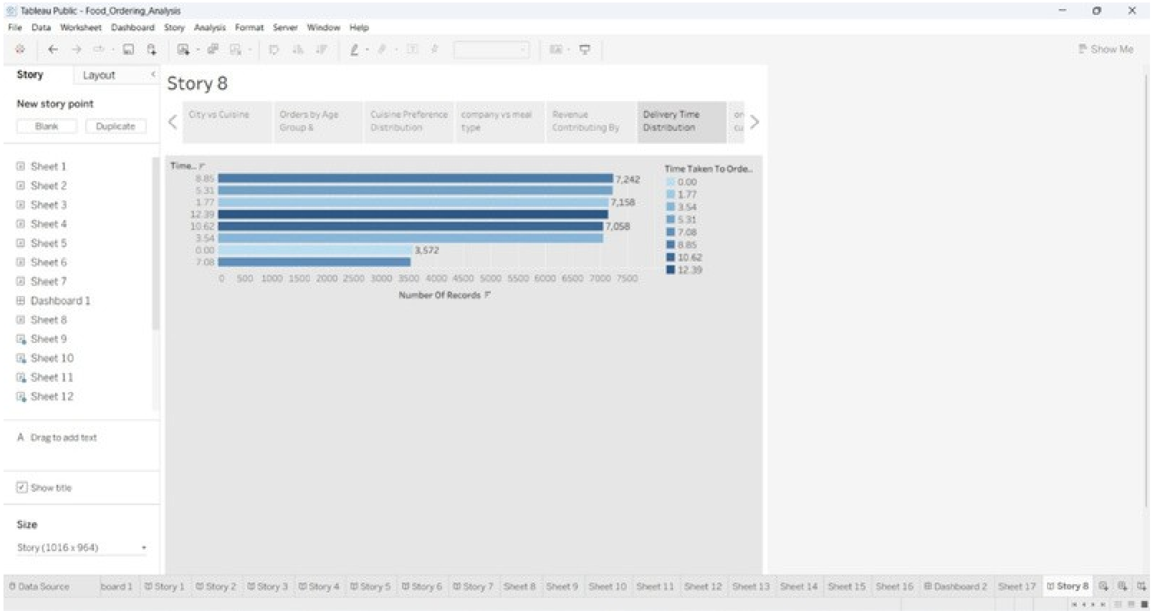
Screenshot 5: Company & Meal Type Orders



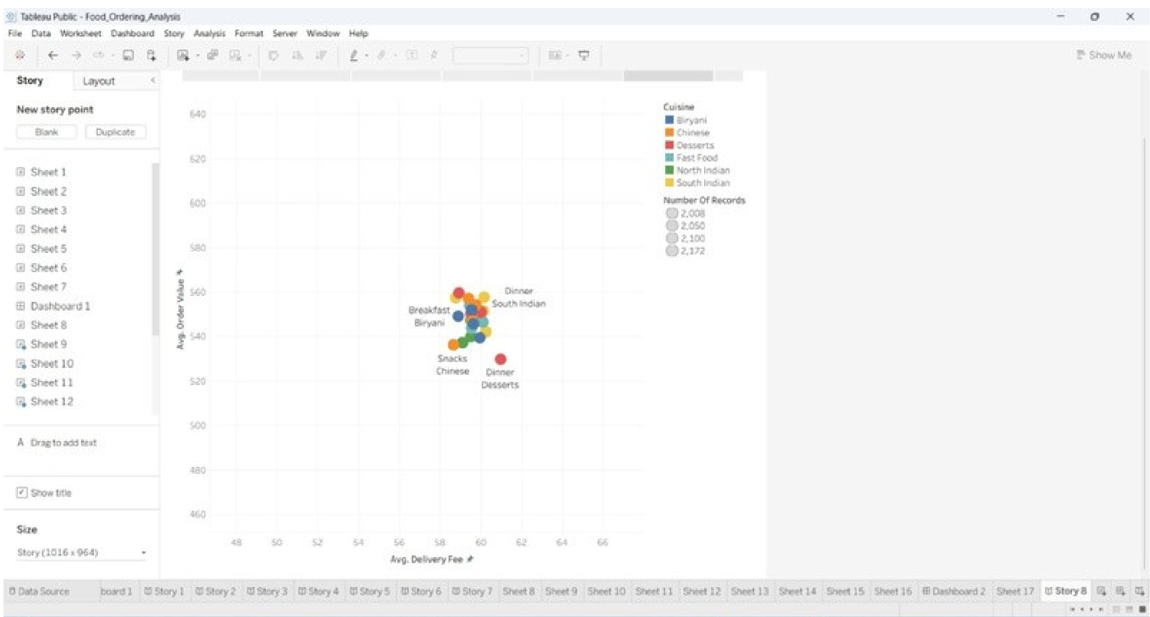
Screenshot 6: Revenue Contributing by Meal Type



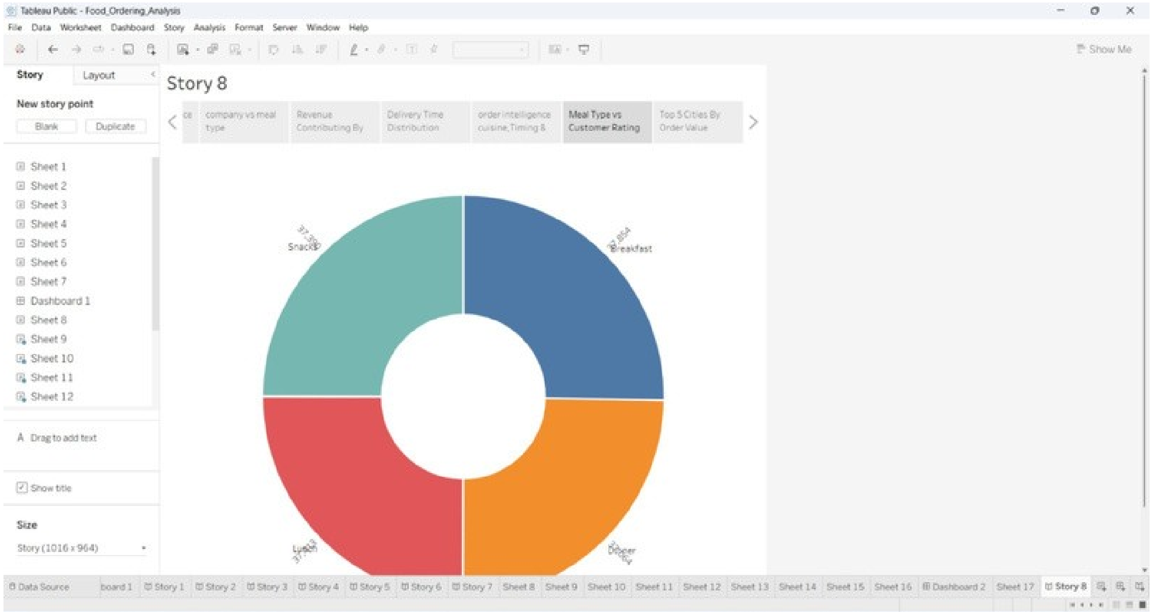
Screenshot 7: Delivery Time Distribution



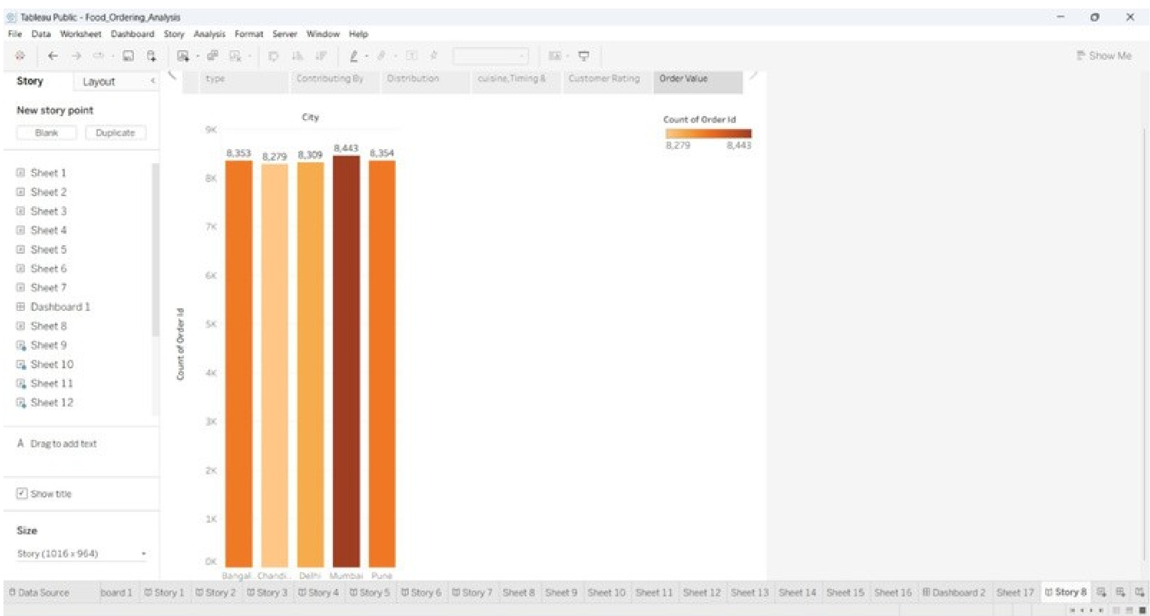
Screenshot 8: Order Intelligence: Cuisine, Timing & Spend



Screenshot 9: Meal Type vs Customer Rating



Screenshot 10: Top 5 Cities by Order Volume



8. Advantages & Disadvantages

Advantages

- Turns raw, hard-to-interpret order data into clear, interactive visual insights.
- Gives Business Analysts and Operations Managers a shared, filterable view of the same data.
- Surfaces concrete, actionable findings — top cities, high-demand cuisines, peak mealtimes, delivery delays — rather than raw numbers.
- Built entirely on Tableau, so it can be extended without custom software development.

Disadvantages

- Insights are only as good as the underlying dataset; missing or inconsistent fields limit accuracy.
- Tableau dashboards are descriptive rather than predictive — they show what happened, not what will happen next.
- Subscription & Loyalty Programs analysis was de-prioritised, so loyalty-driven behaviour is only partially captured.

9. Conclusion

This project shows how a structured, Tableau-based analysis can turn everyday food-ordering data into insight that Business Analysts, Operations Managers, and customer-facing teams can act on. By grounding the analysis in real problem statements and an empathy map of the primary persona, the resulting dashboards stay focused on the factors that most affect ordering behaviour — timing, price sensitivity, delivery performance, and cuisine preference — rather than on data for its own sake.

10. Future Scope

- Extend the analysis to Subscription & Loyalty Programs once more historical data is available.
- Add predictive elements (e.g. demand forecasting by city or cuisine) alongside the current descriptive dashboards.
- Automate the data-refresh pipeline so dashboards update on a schedule rather than through manual re-import.
- Incorporate customer review / sentiment data to enrich the Decision-Influencing Factors analysis.

11. Appendix

Source Code (if any): Not applicable — this project is delivered as Tableau workbooks rather than custom application code.

Dataset Link: OneDrive dataset link

GitHub & Project Demo Link: GitHub repository

Tableau Public Dashboard Link: Dashboard 2

Tableau Public Story Link: Story 8